LOPO Results

Leave-one-participant-out performance across six held-out participants.

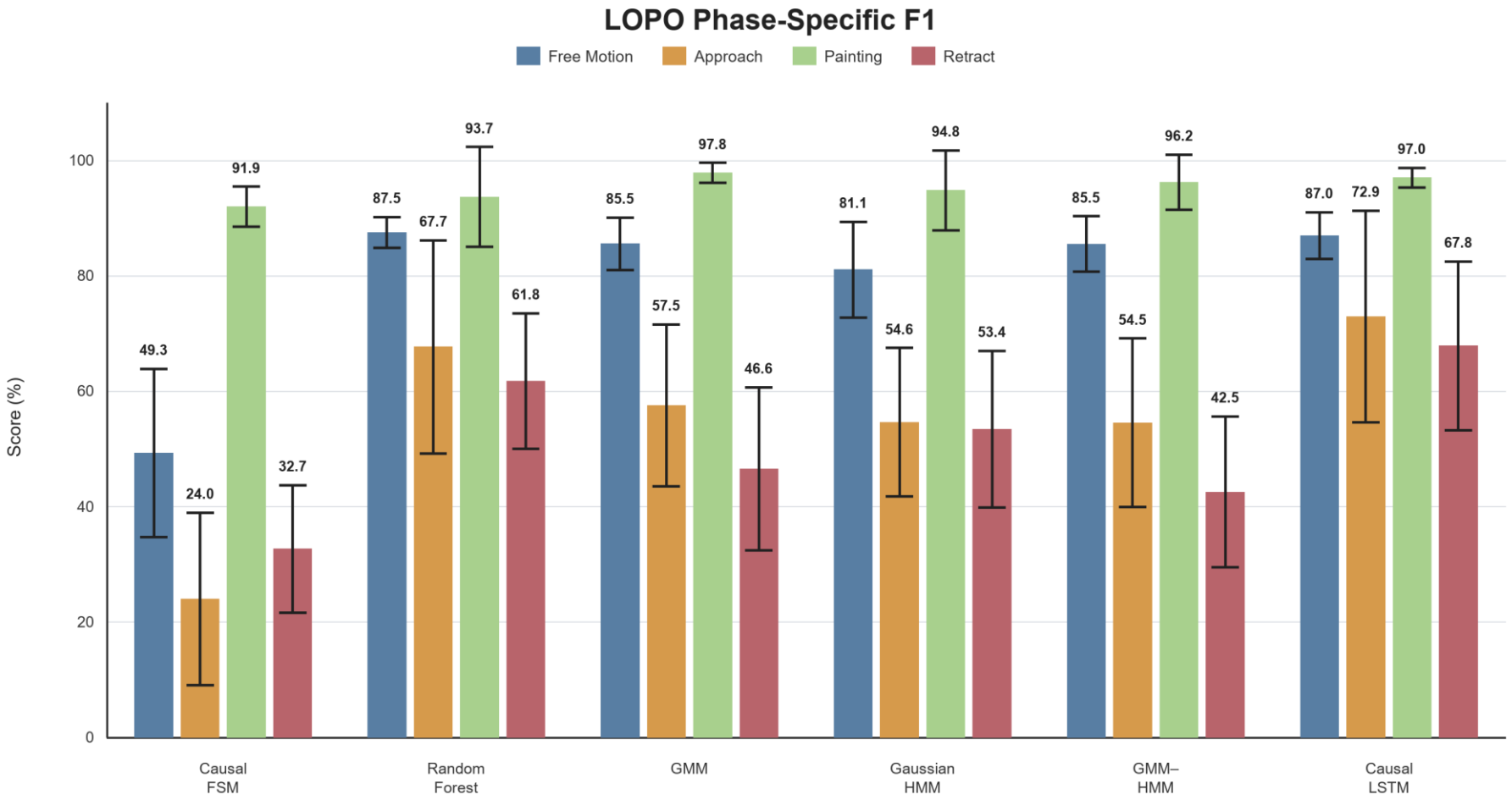


Figure 1. Phase-specific F1 scores. Bars represent means and error bars represent ±1 SD across six LOPO folds.

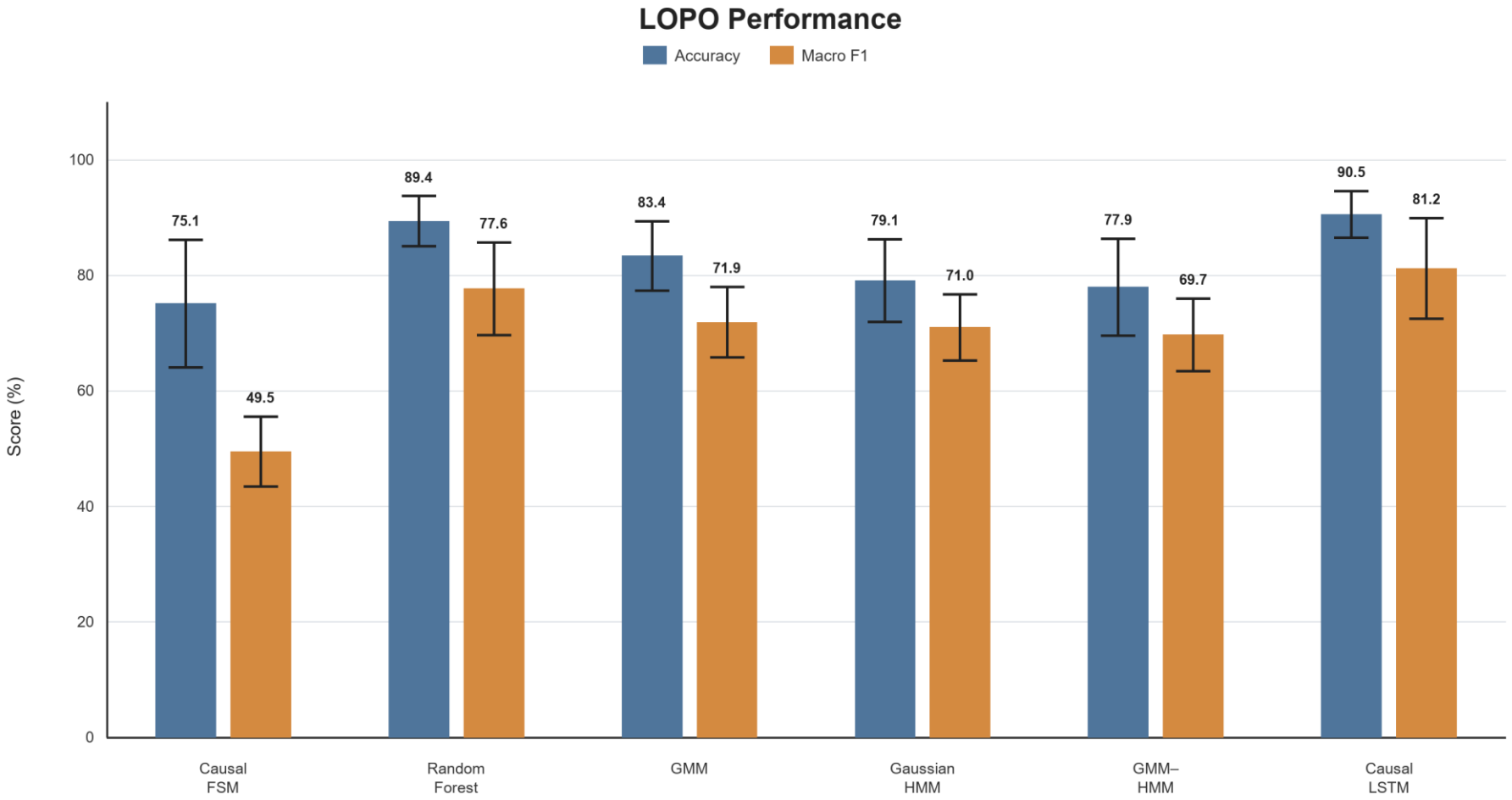


Figure 2. Overall LOPO performance. Bars represent means and error bars represent ±1 SD across six LOPO folds.

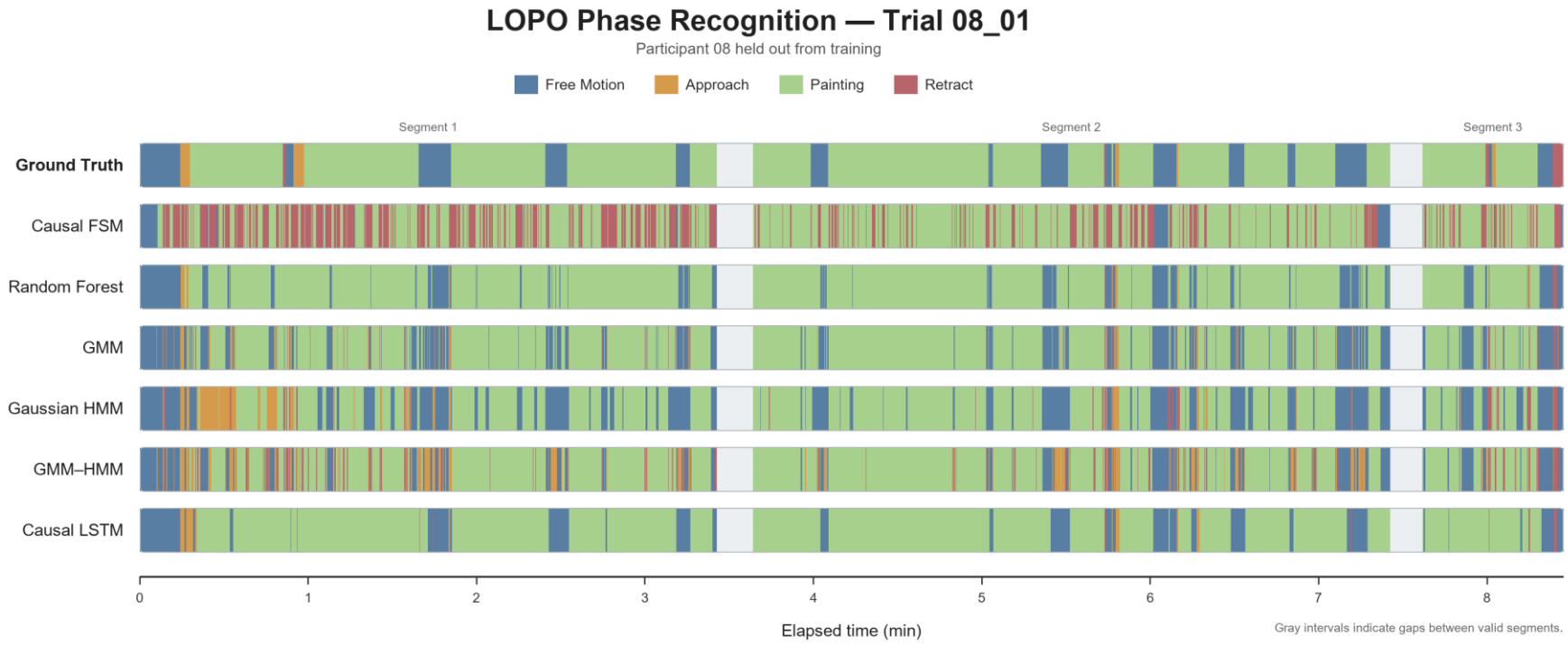


Figure 3. Framewise causal phase predictions for trial 08_01. Participant 08 was excluded from training; gray intervals indicate gaps between valid segments.

Table 1

LOPO Classification and Transition Results Within a ±3-s Tolerance

Model	Overall		Phase-specific F1				Transition
	Accuracy	Macro F1	Free Motion	Approach	Painting	Retract	F1
Causal FSM	75.1 (11.0)	49.5 (6.1)	49.3 (14.6)	24.0 (15.0)	91.9 (3.5)	32.7 (11.0)	6.6 (2.3)
Random Forest	89.4 (4.3)	77.6 (8.0)	87.5 (2.7)	67.7 (18.5)	93.7 (8.6)	61.8 (11.7)	18.3 (5.9)
GMM	83.4 (6.0)	71.9 (6.1)	85.5 (4.5)	57.5 (14.1)	97.8 (1.7)	46.6 (14.1)	10.9 (3.6)
Gaussian HMM	79.1 (7.2)	71.0 (5.7)	81.1 (8.3)	54.6 (12.9)	94.8 (6.9)	53.4 (13.6)	19.4 (1.8)
GMM–HMM	77.9 (8.4)	69.7 (6.3)	85.5 (4.8)	54.5 (14.6)	96.2 (4.8)	42.5 (13.1)	15.6 (6.1)
Causal LSTM	90.5 (4.0)	81.2 (8.7)	87.0 (4.0)	72.9 (18.4)	97.0 (1.7)	67.8 (14.6)	38.9 (10.5)

Note. Values are percentages and are reported as M (SD) across six held-out participants. Classification and transition scores were calculated using a ±3-s tolerance. LOPO = leave-one-participant-out; M = mean; SD = standard deviation.